Recent Advances in Deep Learning - Spring 2026

Final Examination

Total Marks: 100 Time Allowed: 120 minutes

Part I: True/False Questions (10 questions, 1 mark each, Total: 10 marks)

Please indicate whether the following statements are true or false. Check (✓) for True, and cross (×) for False.

1. In Backpropagation, the chain rule is the core mathematical tool for computing the gradients of weights in each layer of a deep network. ( )
2. When updating parameters, the Adam optimizer only uses the first moment (Momentum) estimate of the gradient and does not introduce a second moment estimate. ( )
3. In the Self-Attention computation of the Transformer architecture, the main purpose of dividing the dot product result by the scaling factor $\sqrt{d_k}$ is to prevent the Softmax function from entering regions with vanishing gradients. ( )
4. For a Multi-Layer Perceptron (MLP), as long as the depth of hidden layers and the number of neurons are continuously increased, the model's performance on the test set will definitely improve. ( )
5. The ReLU activation function has a derivative of zero when the input value is less than zero; this characteristic may lead to the "Dying ReLU" phenomenon during training. ( )
6. The Transformer completely abandons traditional RNN and CNN structures, relying solely on attention mechanisms to establish global sequence dependencies. ( )
7. Stochastic Gradient Descent (SGD) uses the entire training set data to calculate the gradient and update parameters at each iteration. ( )
8. In the self-attention mechanism, the role of the Value matrix (V) is to calculate the correlation score between different tokens in the input sequence. ( )
9. The Skip Connection in Residual Networks (ResNet) can effectively alleviate the problems of exploding and vanishing gradients in deep neural networks. ( )
10. Layer Normalization is widely used in the Transformer architecture, and it is calculated by normalizing across all feature dimensions for a single sample. ( )

Part II: Multiple Choice Questions (20 questions, 2 marks each, Total: 40 marks)

For each question below, there is only one correct answer. Please select the most appropriate option.

1. Assume a hidden layer of a Multi-Layer Perceptron (MLP) has N input nodes and M output nodes. Ignoring the bias term, the dimension of the weight matrix for this layer is:

   A. $N \times N$

   B. $M \times M$

   C. $N \times M$

   D. $N + M$

2. Which of the following optimization algorithms combines adaptive learning rates and the Momentum mechanism?

   A. SGD

   B. AdaGrad

   C. RMSProp

   D. Adam

3. The formula for the standard Scaled Dot-Product Attention in the Transformer is:

   A. $\text{Attention}(Q, K, V) = \text{Softmax}(QK^T)V$

   B. $\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$

   C. $\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QV^T}{\sqrt{d_k}}\right)K$

   D. $\text{Attention}(Q, K, V) = \text{Softmax}(QK^T \cdot \sqrt{d_k})V$

4. During backpropagation, if continuously multiplied gradient values are all less than 1, which of the following problems is most likely to occur?

   A. Exploding gradients

   B. Vanishing gradients

   C. Internal covariate shift

   D. Overfitting

5. What is the primary purpose of using Dropout in deep learning?

   A. To accelerate model convergence

   B. To reduce computational complexity

   C. To prevent model overfitting

   D. To solve the vanishing gradient problem

6. When training very deep neural networks, which activation function is generally recommended to replace the traditional Sigmoid function to improve gradient flow?

   A. Tanh

B. Step Function

C. ReLU and its variants

D. Softmax

7. Regarding the Multi-Head Attention mechanism in the Transformer, which of the following descriptions is INCORRECT?

   A. It allows the model to jointly attend to information from different representation subspaces at different positions.

   B. Its computational complexity grows exponentially compared to single-head attention.

   C. It concatenates the outputs of multiple attention heads and applies a linear transformation to get the final result.

   D. It helps capture richer contextual features.

8. Given a sequence length of L and a feature dimension of D, the computational time complexity (excluding batch processing) of the standard self-attention mechanism is approximately:

   A. $O(L \cdot D^2)$

   B. $O(L^2 \cdot D)$

   C. $O(L \cdot D)$

   D. $O(L^2 \cdot D^2)$

9. The main physical intuition behind introducing the Momentum term in the SGD optimizer is:

   A. Simulating the temperature annealing process.

   B. Simulating the inertial motion of an object under friction, reducing update oscillations.

   C. Assigning learning rates of different magnitudes to different parameters.

   D. Completely eliminating the impact of the learning rate.

10. In classification tasks, the most commonly used combination of activation function in the output layer and corresponding loss function is:

    A. ReLU, Mean Squared Error (MSE)

    B. Sigmoid, Cross-Entropy Loss

    C. Softmax, Cross-Entropy Loss

    D. Tanh, KL-Divergence

11. What mechanism did the Transformer introduce to allow the model to perceive the order of the input sequence?

    A. Hidden state transfer

    B. Positional Encoding

    C. Convolutional layers to extract local features

    D. Sequence reversal

12. Which normalization method is most suitable for handling variable-length inputs in sequence models like RNNs or Transformers?

    A. Batch Normalization
    B. Layer Normalization
    C. Instance Normalization
    D. Group Normalization

13. How many linear transformations does the Feed-Forward Network (FFN) contain in each encoder layer of the Transformer?

    A. 1
    B. 2, with an activation function in between
    C. 3
    D. The number is determined by the number of attention heads

14. Which of the following is NOT a common method to prevent neural network overfitting?

    A. L2 Regularization (Weight Decay)
    B. Increasing training data
    C. Decreasing Batch Size
    D. Early Stopping

15. Compared to L2 regularization, what unique geometric property and optimization effect does L1 regularization have?

    A. It causes model weights to tend towards a sparse matrix (many weights become 0).
    B. It makes the absolute values of all weights exactly equal.
    C. It turns the gradient calculation into a non-convex optimization problem.
    D. It prevents exploding gradients more effectively.

16. When calculating self-attention scores, how is the Query matrix obtained?

    A. Randomly initialized and generated.
    B. By multiplying the output of the previous layer by a learnable weight matrix W^Q.
    C. By directly using the input vector as the Query.
    D. By applying average pooling to the input.

17. Which of the following learning rate decay strategies (Learning Rate Scheduler) is most common in Transformer pre-training?

    A. Constant learning rate
    B. Step Decay
    C. Warmup with Cosine Decay
    D. Linear increase

18. The Decoder module of the Transformer contains a special attention mechanism to prevent information leakage. It is:

A. Cross-Attention
B. Masked Self-Attention
C. Sparse Attention
D. Local Attention

19. Regarding computational graphs and backpropagation, which of the following statements is correct?

    A. Forward propagation calculates the gradient, and backpropagation calculates the loss.
    B. Every differentiable operation can be represented as a node on the computational graph, and backpropagation is executing the chain rule on the graph.
    C. The 'autograd' in PyTorch can only calculate the derivative of a scalar with respect to a vector and cannot handle matrix multiplication.
    D. Once a computational graph is statically built, its shape cannot be changed during runtime.

20. Which of the following is the key first step operation when the Transformer model processes images (e.g., Vision Transformer, ViT)?

    A. Applying 3D convolution to the image.
    B. Splitting the image into multiple non-overlapping Patches and flattening them into a sequence.
    C. Using a fully connected layer to output pixel-level labels.
    D. Directly calculating global attention on all RGB pixels.

Part III: Short Answer & Essay Questions (Open-end, 5 questions, 10 marks each, Total: 50 marks)

1. **Multilayer Perceptron and Backpropagation (10 marks)**
   Consider a simple two-layer MLP: $h = \sigma(W_1 x + b_1)$, $y = W_2 h + b_2$, where σ is the activation function, x is the input, and y is the predicted output. Assume the loss function is Mean Squared Error $L = \frac{1}{2}\|y - t\|^2$ (where t is the target value). Based on the chain rule of calculus, derive the expression for the gradient of the loss function L with respect to the weight matrix W_1, i.e., $\frac{\partial L}{\partial W_1}$.

2. **Analysis of Neural Network Optimizers (10 marks)**
   In deep learning training, the Adam optimizer has achieved immense success compared to standard Stochastic Gradient Descent (SGD). Briefly describe the core idea of the Adam optimizer. How does it combine Momentum and adaptive learning rates (the RMSProp concept)? Furthermore, point out why researchers sometimes still prefer SGD with momentum over Adam for certain specific tasks.

3. **Detailed Explanation of Attention Mechanism Computation (10 marks)**
   Provide a detailed explanation of the calculation process for "Scaled Dot-Product Self-Attention" in the Transformer. Requirements: (1) Specify how Q, K, and V are generated; (2) Explain why it is necessary to divide by $\sqrt{d_k}$ after calculating QK^T; (3) Briefly describe the additional advantages brought by the Multi-Head mechanism.

4. **Architecture Design and Gradient Propagation Analysis (10 marks)**
   The "Vanishing Gradient" is one of the major bottlenecks restricting the training of deep neural networks. Please explain the essential mathematical cause of vanishing gradients. Meanwhile, discuss how the "Skip Connection" (or Residual Connection) in Deep Residual Networks (ResNet) effectively alleviates this problem from a mathematical structure perspective.

5. **Recent Advances and Bottlenecks in Deep Learning (10 marks)**
   Based on the theoretical knowledge you learned in the *Recent Advances in Deep Learning* course, discuss the significant advantages (e.g., parallelization capability, extraction of long-range dependencies) of the Transformer architecture in long sequence modeling compared to traditional Long Short-Term Memory (LSTM) networks. Additionally, discuss the core computational bottleneck faced by the standard Transformer architecture when processing extremely long sequences (e.g., 100,000-level tokens), and list one potential approach from academia to mitigate this bottleneck.